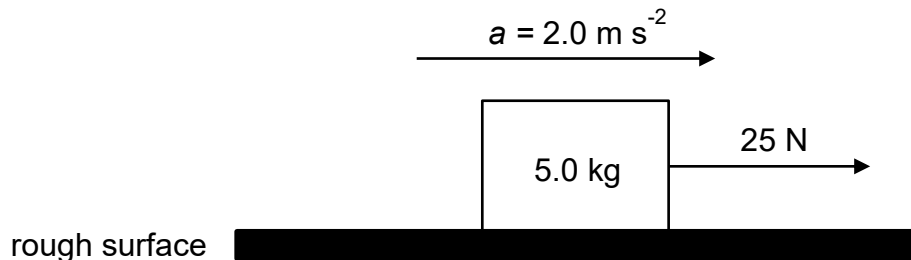


- 8** A block is pulled with a constant force of 25 N on a rough horizontal surface. It accelerates with a constant acceleration of 2.0 m s^{-2} .



What percentage of the power supplied by the 25 N force is lost as heat due to the rough surface?

- | | | | | | | | |
|----------|-----|----------|------|----------|------|----------|------|
| A | 0 % | B | 33 % | C | 60 % | D | 67 % |
|----------|-----|----------|------|----------|------|----------|------|

Answer: C

Net force acting on the block = $F - f = ma$

$$25 - f = (5.0)(2.0)$$

$$f = 15 \text{ N}$$

$$\text{Power lost to friction} = \frac{fv}{Fv} \times 100\% = \frac{15}{25} \times 100\% = 60\%$$

- 9** The diagram shows an object rotating in a vertical circle in the drum of the spin drier which